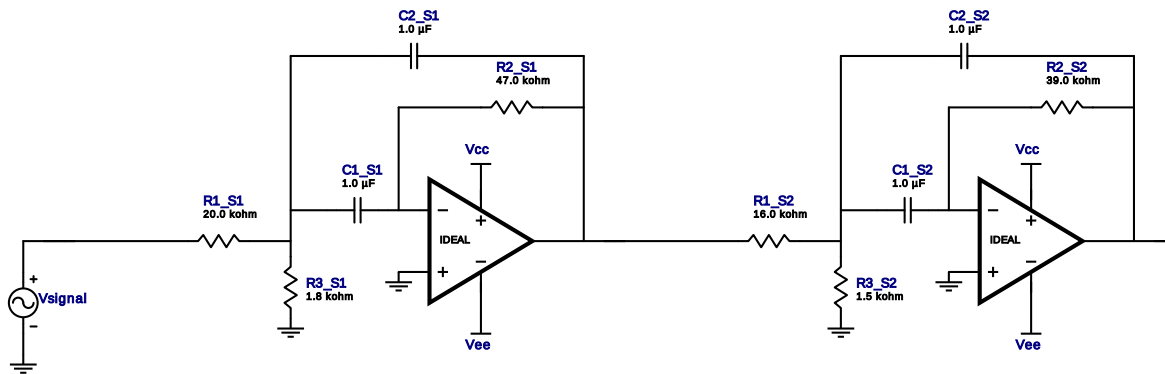


Filter Design Report

Design : Bandpass Filter - 4th order Bessel
Design ID: 58

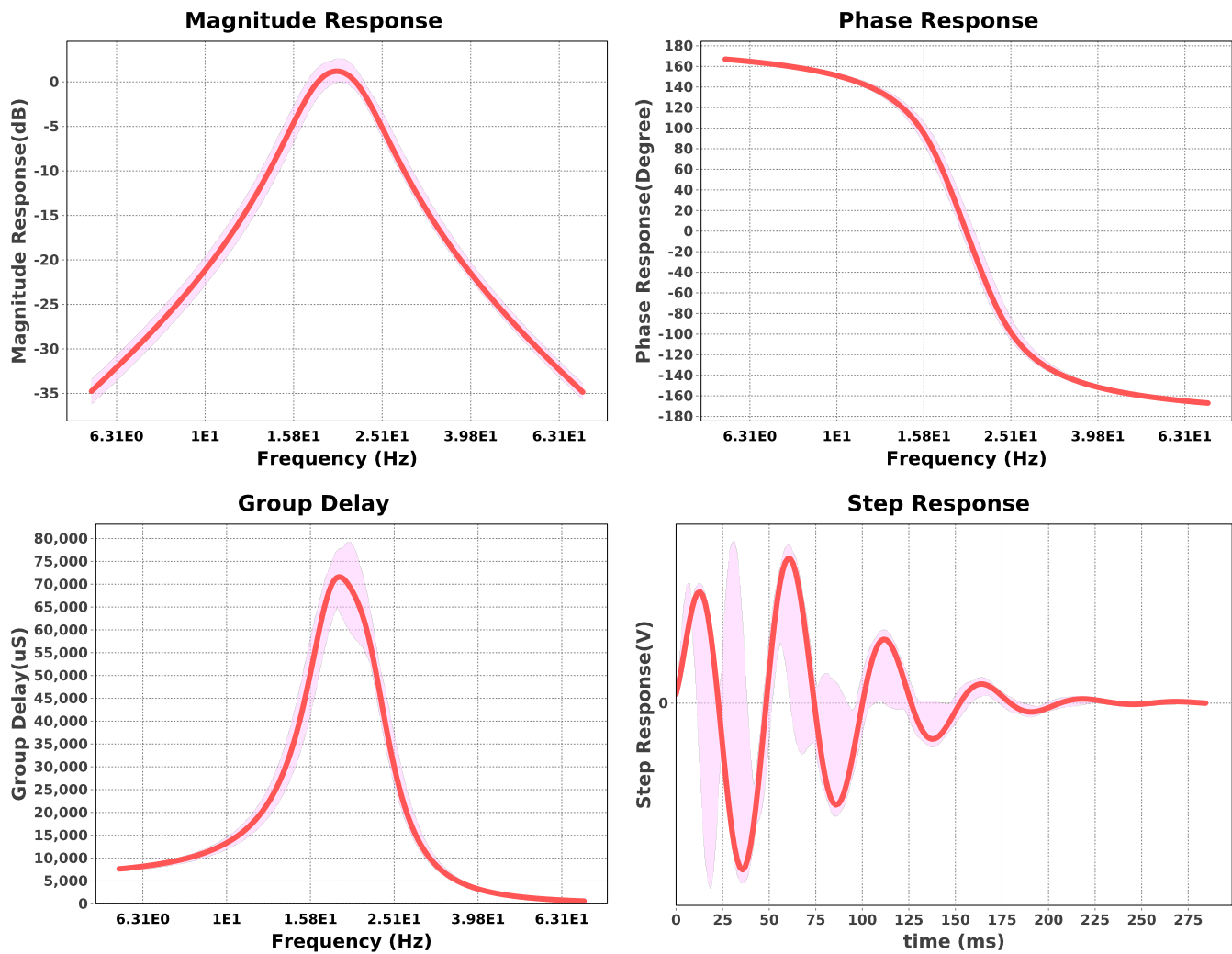


Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S1	Texas Instruments Inc.	IDEAL	GbwTyp= 0MHz VccMax= 0V VccMin= 0V	1
2.	A1_S2	Texas Instruments Inc.	IDEAL	GbwTyp= 0MHz VccMax= 0V VccMin= 0V	1
3.	C1_S1	Generic	Ideal	Cap= 1.0 uF Tolerance= 2.0 %	1
4.	C1_S2	Generic	Ideal	Cap= 1.0 uF Tolerance= 2.0 %	1
5.	C2_S1	Generic	Ideal	Cap= 1.0 uF Tolerance= 2.0 %	1
6.	C2_S2	Generic	Ideal	Cap= 1.0 uF Tolerance= 2.0 %	1
7.	R1_S1	Generic	Ideal	Res= 20000.0ohm Tolerance= 5%	1
8.	R1_S2	Generic	Ideal	Res= 16000.0ohm Tolerance= 5%	1
9.	R2_S1	Generic	Ideal	Res= 47000.0ohm Tolerance= 5%	1
10.	R2_S2	Generic	Ideal	Res= 39000.0ohm Tolerance= 5%	1
11.	R3_S1	Generic	Ideal	Res= 1800.0ohm Tolerance= 5%	1
12.	R3_S2	Generic	Ideal	Res= 1500.0ohm Tolerance= 5%	1

Sensitivity Analysis

#	Name	Series	Tolerance
1.	Cap	E48	2%
2.	Res	E24	5%



Design Inputs

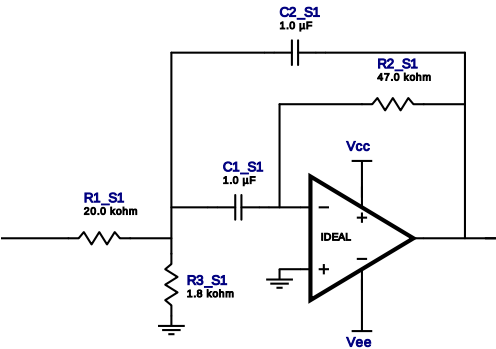
#	Name	Value	Description
1.	FilterType	bandpass	
2.	FilterResponse	Bessel	
3.	FilterOrder	4.0	
4.	FilterTopology	Multiple Feedback	
5.	NumberOfStages	2.0	
6.	CenterFrequency	20.0	
7.	StopbandAttenuation	-42.865	
8.	PassbandBandwidth	6.67	
9.	StopbandBandwidth	100.0	
10.	Gain	1.0	
11.	DualSupply	+/-5.00 V	Power supply(s) to active chips
12.	ResistorTolerance	E24	Resistor series - 5% Passive resistor tolerance
13.	CapacitorTolerance	E48	Capacitor series - 2% Passive capacitor tolerance

Design Assistance

1. **IDEAL** Product Folder : <http://www.ti.com/product/IDEAL> : contains the data sheet and other resources.

Filter Stage :1

Cutoff Frequency 18.065 Hz
Min GBW Reqd 5.702 kHz
Stage Gain 1.175 V/V
Stage Q 2.667
Stage Topology Multiple Feedback

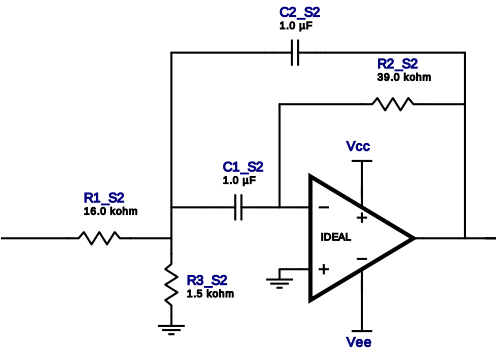


Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S1	Texas Instruments Inc.	IDEAL	GbwTyp= 0MHz VccMax= 0V VccMin= 0V	1
2.	C1_S1	Generic	Ideal	Cap= 1.0 uF Tolerance= 2.0 %	1
3.	C2_S1	Generic	Ideal	Cap= 1.0 uF Tolerance= 2.0 %	1
4.	R1_S1	Generic	Ideal	Res= 20000.0ohm Tolerance= 5%	1
5.	R2_S1	Generic	Ideal	Res= 47000.0ohm Tolerance= 5%	1
6.	R3_S1	Generic	Ideal	Res= 1800.0ohm Tolerance= 5%	1

Filter Stage :2

Cutoff Frequency 21.762 Hz
Min GBW Reqd 7.074 kHz
Stage Gain 1.219 V/V
Stage Q 2.666
Stage Topology Multiple Feedback



Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S2	Texas Instruments Inc.	IDEAL	GbwTyp= 0MHz VccMax= 0V VccMin= 0V	1
2.	C1_S2	Generic	Ideal	Cap= 1.0 uF Tolerance= 2.0 %	1
3.	C2_S2	Generic	Ideal	Cap= 1.0 uF Tolerance= 2.0 %	1
4.	R1_S2	Generic	Ideal	Res= 16000.0ohm Tolerance= 5%	1
5.	R2_S2	Generic	Ideal	Res= 39000.0ohm Tolerance= 5%	1
6.	R3_S2	Generic	Ideal	Res= 1500.0ohm Tolerance= 5%	1

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